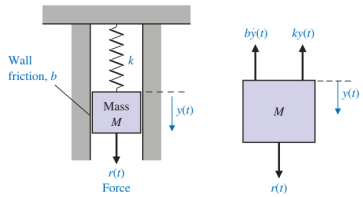
Modelos matemáticos de sistemas físicos

1

Reconsideremos el sistema mecánico…



$$M\frac{d^2y(t)}{dt^2} + b\frac{dy(t)}{dt} + ky(t) = r(t)$$ ⇩ Transformada Laplace

$$M\left(s^2Y(s) - sy(0^-) - \frac{dy}{dt}\,(0^-)\right) + b\left(sY(s) - y(0^-)\right) + kY(s) = R(s)$$

SI ⇩ $r(t) = 0, \quad \text{and} \quad y(0^-) = y_0, \quad \text{and} \quad \left.\frac{dy}{dt}\right|_{t=0^-} = 0,$

$$Ms^2Y(s) - Msy_0 + bsY(s) - by_0 + kY(s) = 0.$$

⇩

$$Y(s) = \frac{(Ms+b)\,y_0}{Ms^2 + bs + k} = \frac{p(s)}{q(s)}.$$ Si se propone como ejemplo ⇩ $k/M = 2$, $b/M = 3$

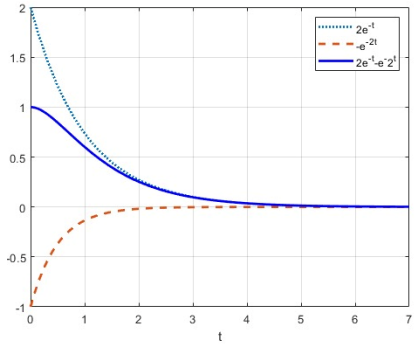
$$Y(s) = \frac{(s+3)y_0}{(s+1)(s+2)}.$$

Fracciones parciales ⇩

$$Y(s) = \frac{k_1}{s+1} + \frac{k_2}{s+2}$$

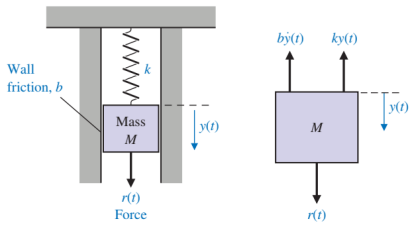
$$y(t) = \mathcal{L}^{-1}\left\{\frac{2}{s+1}\right\} + \mathcal{L}^{-1}\left\{\frac{-1}{s+2}\right\}$$

$$y(t) = 2e^{-t} - 1e^{-2t}.$$



2

Siguiendo con el mismo sistema mecánico, pero con diferente comportamiento



Se había obtenido ⇨ $$Y(s) = \frac{(Ms+b)\,y_0}{Ms^2+bs+k} = \frac{p(s)}{q(s)}.$$

$$Y(s) = \frac{(s+b/M)\,y_0}{s^2+(b/M)s+k/M} = \frac{(s+2\zeta\omega_n)\,y_0}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$\omega_n = \sqrt{k/M}$$

$$\zeta = b/\left(2\sqrt{kM}\right)$$

ζ ⇨ Factor de amortiguamiento ω_n ⇨ Frecuencia natural

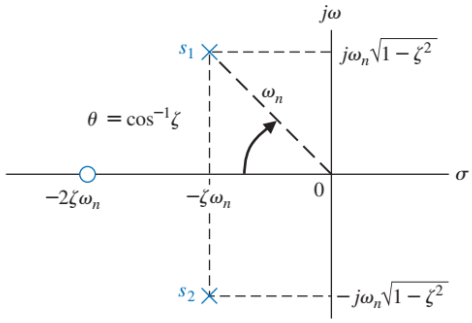
Ahora supongamos que las raíces son complejas

$$s_1,\ s_2 = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$$

Si $\zeta < 1$ El sistema es subamortiguado

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

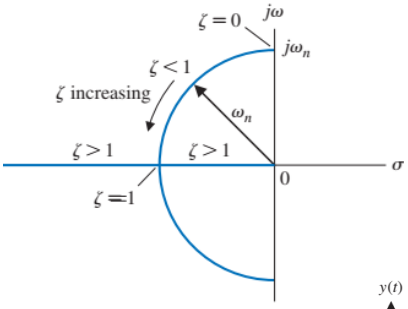
Ubicación de ceros y polos



3

Si se varía sita manteniendo Wn constante

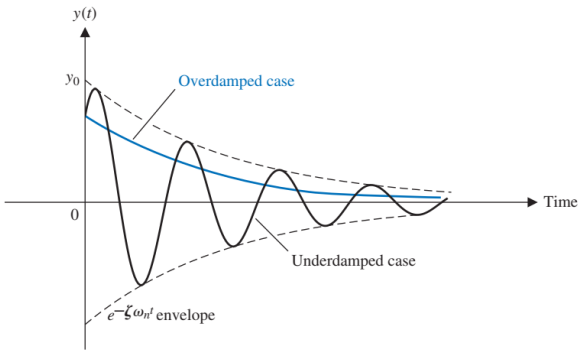
Haciendo la anti transformada de la salida y denotando $\beta = \sqrt{1-\zeta^2}$



$$y(t) = k_1e^{s_1t} + k_2e^{s_2t}$$

$$= \frac{y_0}{2\sqrt{1-\zeta^2}}\left(e^{j(\theta-\pi/2)}e^{-\zeta\omega_nt}e^{j\omega_n\beta t} + e^{j(\pi/2-\theta)}e^{-\zeta\omega_nt}e^{-j\omega_n\beta t}\right)$$

$$= \frac{y_0}{\sqrt{1-\zeta^2}}\,e^{-\zeta\omega_nt}\,\sin\left(\omega_n\sqrt{1-\zeta^2}\,t+\theta\right).$$

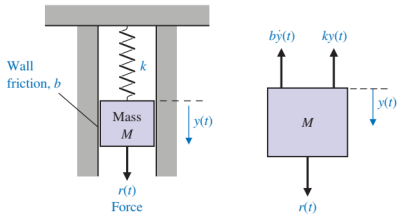


4

Función de transferencia

Definición: para un sistema LTI, la función de transferencia es la relación entre la transformada de Laplace de la salida y la de la entrada. Se deben considerar a todas las condiciones iniciales iguales a cero.

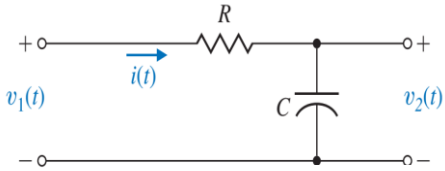
Volviendo al caso mecánico y planteando para condiciones iniciales cero



$$Ms^2Y(s) + bsY(s) + kY(s) = R(s).$$

$$G(s) = \frac{Y(s)}{R(s)} = \frac{1}{Ms^2 + bs + k}.$$

Otro caso:



$$V_1(s) = \left(R + \frac{1}{Cs}\right)I(s) \qquad V_2(s) = I(s)\left(\frac{1}{Cs}\right).$$

$$V_2(s) = \frac{(1/Cs)V_1(s)}{R + 1/Cs}.$$

$$G(s) = \frac{V_2(s)}{V_1(s)} = \frac{1}{RCs + 1} = \frac{1}{\tau s + 1} = \frac{1/\tau}{s + 1/\tau},$$

$\tau = RC$ ⇨ **Constante de tiempo**

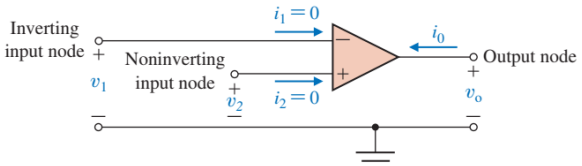
Más directo: considerar al sistema como divisor de tensión

$$\frac{V_2(s)}{V_1(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$

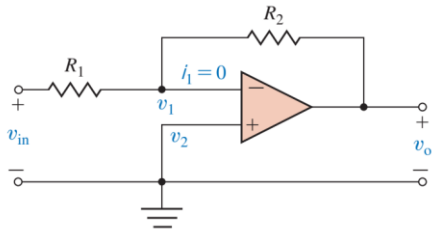
Trabajando esta expresión se llega a lo mismo que antes

5

Amplificador Inversor con amplificador operacional.



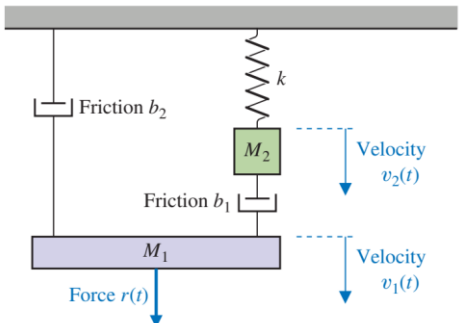
La relación básica de un operacional es ⇨ $v_0 = K(v_2 - v_1) = -K(v_1 - v_2),$



$$\frac{v_1 - v_{in}}{R_1} + \frac{v_1 - v_0}{R_2} = 0.$$ Debido a que V2=V1 y V2=0

$$-\frac{v_{in}}{R_1} - \frac{v_0}{R_2} = 0, \quad \Rightarrow \quad \frac{v_0}{v_{in}} = -\frac{R_2}{R_1}.$$

Sistema mecánico más complejo



$$M_1sV_1(s) + (b_1 + b_2)V_1(s) - b_1V_2(s) = R(s),$$

$$M_2sV_2(s) + b_1V_2((s) - V_1(s)) + k\frac{V_2(s)}{s} = 0.$$

Reordenando las expresiones

$$(M_1s + (b_1 + b_2))V_1(s) + (-b_1)V_2(s) = R(s)$$

$$(-b_1)V_1(s) + \left(M_2s + b_1 + \frac{k}{s}\right)V_2(s) = 0$$

6

Re escribiendo por conveniencia

$$(M_1 s + (b_1 + b_2))V_1(s) + (-b_1)V_2(s) = R(s)$$
$$(-b_1)V_1(s) + \left(M_2 s + b_1 + \frac{k}{s}\right)V_2(s) = 0$$

$$\Rightarrow \begin{bmatrix} M_1 s + b_1 + b_2 & -b_1 \\ -b_1 & M_2 s + b_1 + \frac{k}{s} \end{bmatrix} \begin{bmatrix} V_1(s) \\ V_2(s) \end{bmatrix} = \begin{bmatrix} R(s) \\ 0 \end{bmatrix}$$

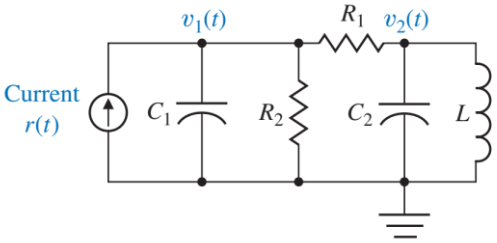
Resolviendo para V1 (considerando que esta es la variable de salida

$$V_1(s) = \frac{(M_2 s + b_1 + k/s)R(s)}{(M_1 s + b_1 + b_2)(M_2 s + b_1 + k/s) - b_1^2} \Rightarrow G(s) = \frac{V_1(s)}{R(s)} = \frac{(M_2 s + b_1 + k/s)}{(M_1 s + b_1 + b_2)(M_2 s + b_1 + k/s) - b_1^2}$$

$$\Rightarrow = \frac{(M_2 s^2 + b_1 s + k)}{(M_1 s + b_1 + b_2)(M_2 s^2 + b_1 s + k) - b_1^2 s}$$

Las ecuaciones serían formalmente similares para el circuito pasivo mostrado, si se hace la siguiente analogía:

$$C_1 = M_1,\ C_2 = M_2,$$
$$L = 1/k,\ R_1 = 1/b_1,$$
$$R_2 = 1/b_2.$$

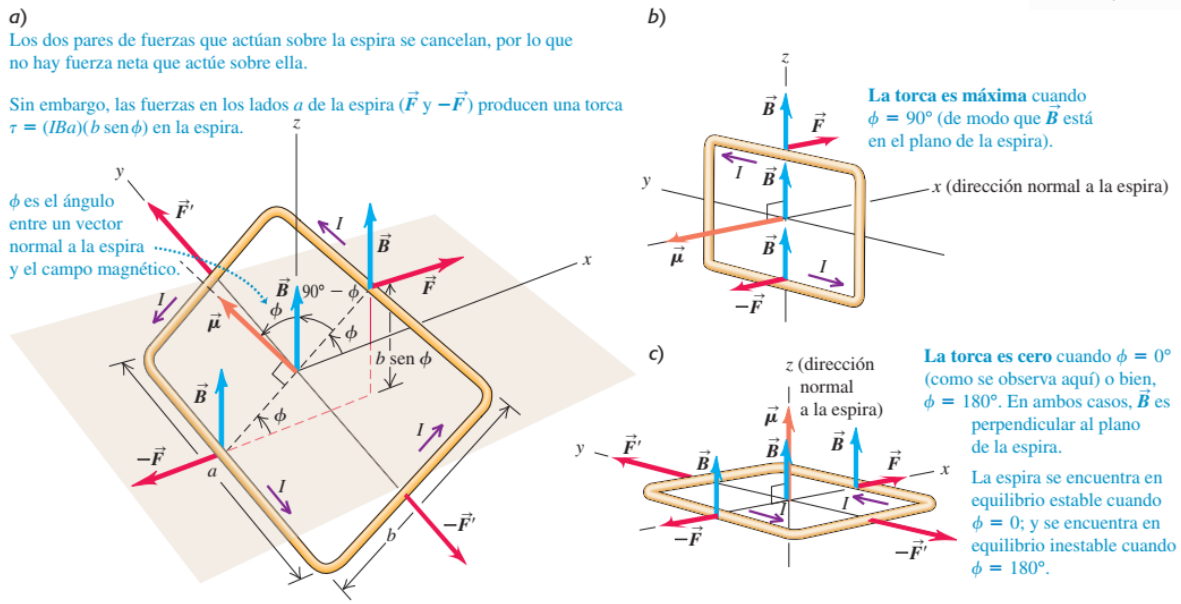


7

Conceptos relacionados a motores de DC

Nota: fuerza experimentada por un conductor con corriente en el seno de un campo magnético

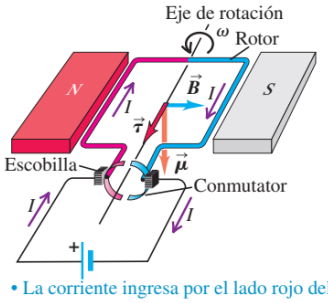
$$F = I(L \times B)$$



8

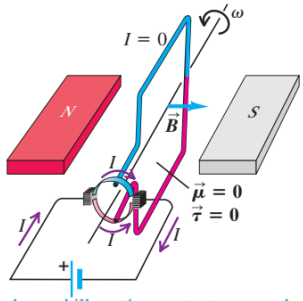
Conceptos relacionados a motores de DC

a) Las escobillas están alineadas con los segmentos del conmutador.



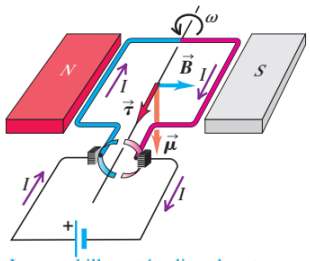
- La corriente ingresa por el lado rojo del rotor y sale por el lado azul.
- Por lo tanto, la torca magnética hace que el rotor gire en sentido.

b) El rotor ha girado 90°.



- Cada escobilla está en contacto con ambos segmentos del conmutador, por lo que la corriente se desvía totalmente del rotor.
- Ninguna torca magnética actúa sobre el rotor.

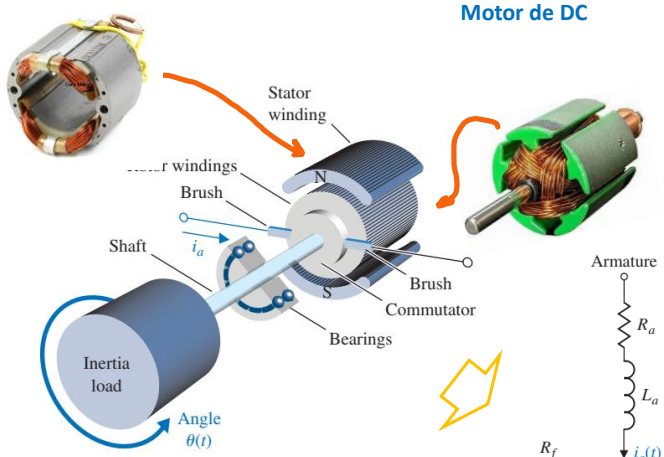
c) El rotor ha girado180°.

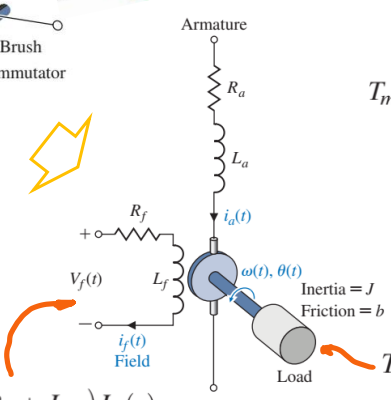


- Las escobillas están alineadas otra vez con los segmentos del conmutador. Esta vez la corriente entra por el lado azul del rotor y sale por el lado rojo.
- Por lo tanto, la torca magnética otra vez ocasiona que el rotor gire en sentido antihorario.

9

Motor de DC





$$V_f(s) = \left(R_f + L_f s\right) I_f(s).$$

Consideremos las siguientes relaciones

$$\phi(t) = K_f i_f(t)$$

$$T_m(t) = K_1 \phi(t) i_a(t) = K_1 K_f i_f(t) i_a(t)$$

Supongamos que se controla el motor por campo (*f*), y se mantiene constante *Ia*.

$$T_m(s) = \left(K_1 K_f I_a\right) I_f(s) = K_m I_f(s)$$

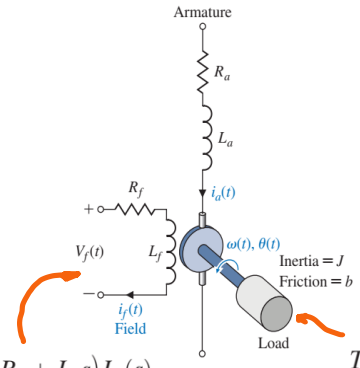
En esta expresión Td es un torque de perturbación. Es para plantear un modelo general. Se lo considera despreciable.

$$T_m(s) = T_L(s) + T_d(s)$$

$$T_L(s) = Js^2\theta(s) + bs\theta(s)$$

10

Motor de DC



$$V_f(s) = (R_f + L_f s) I_f(s).$$

$$T_L(s) = Js^2\theta(s) + bs\theta(s)$$

Reordenando por conveniencia

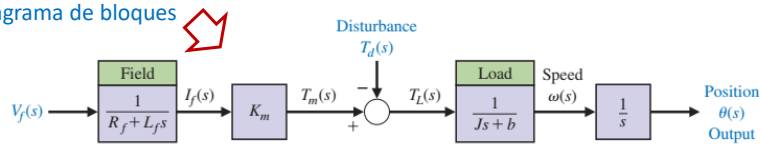
$$T_L(s) = T_m(s) - T_d(s),$$

$$T_m(s) = K_m I_f(s),$$

$$I_f(s) = \frac{V_f(s)}{R_f + L_f s}.$$

$$\frac{\theta(s)}{V_f(s)} = \frac{K_m}{s(Js+b)(L_f s + R_f)} = \frac{K_m/(JL_f)}{s(s+b/J)(s+R_f/L_f)}.$$

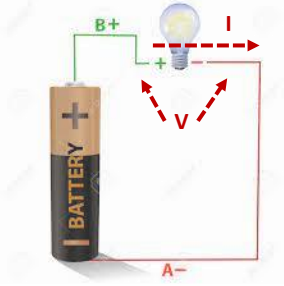
Diagrama de bloques



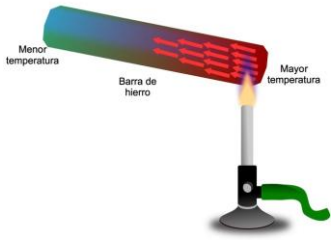
11

Concepto de variables through y across

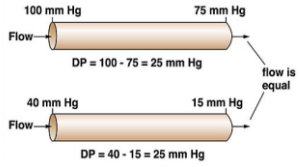
Batería con una lámpara



Diferencia de temperatura y flujo de calor

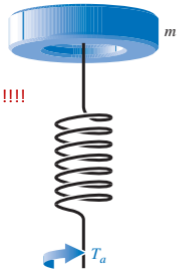


Diferencia de presión y flujo



Torque y velocidad angular

Caso menos intuitivo!!!!!!





Si asumimos que la masa del resorte es despreciable (no tiene momento de inercia)

$$T_a(t) - T_s(t) = 0,$$

$$T_s(t) = T_a(t)$$

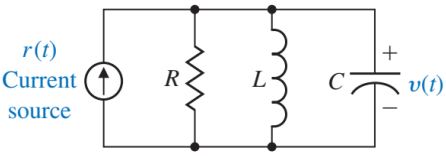
Por otro lado

$$\omega(t) = \omega_s(t) - \omega_a(t).$$

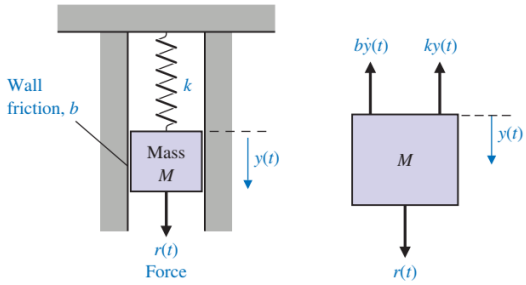
Consecuentemente: *T* es through, *W* es across

12

Profundizando sobre variables through y across



$$\frac{v(t)}{R} + C\frac{dv(t)}{dt} + \frac{1}{L}\int_0^t v(t)\,dt = r(t) \quad \textbf{(1)}$$



$$M\frac{d^2y(t)}{dt^2} + b\frac{dy(t)}{dt} + ky(t) = r(t)$$

Teniendo en cuenta que: $v(t) = \dfrac{dy(t)}{dt}$

$$\textbf{(2)} \quad M\frac{dv(t)}{dt} + bv(t) + k\int_0^t v(t)\,dt = r(t)$$

Comparando (1) y (2) es posible determinar que el voltaje en los circuitos eléctricos (across) se comporta como la velocidad en los sistemas mecánicos de traslación.
Por otro lado, la corriente (through) se comporta como la fuerza.
De la comparación del planteamiento de las ecuaciones diferenciales para cada sistema, se pueden determinar cuáles son las variables through y cuáles las across.

13

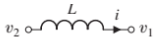
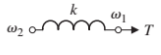
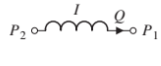
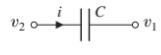
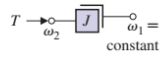
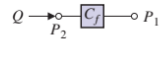
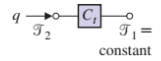
Variables through y across para diversos sistemas físicos

Table 2.1 Summary of Through- and Across-Variables for Physical Systems

System	Variable Through Element	Integrated Through-Variable	Variable Across Element	Integrated Across-Variable
Electrical	Current, i	Charge, q	Voltage difference, v_{21}	Flux linkage, λ_{21}
Mechanical translational	Force, F	Translational momentum, P	Velocity difference, v_{21}	Displacement difference, y_{21}
Mechanical rotational	Torque, T	Angular momentum, h	Angular velocity difference, ω_{21}	Angular displacement difference, θ_{21}
Fluid	Fluid volumetric rate of flow, Q	Volume, V	Pressure difference, P_{21}	Pressure momentum, γ_{21}
Thermal	Heat flow rate, q	Heat energy, H	Temperature difference, $\mathcal{T}_{21}$	

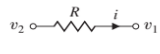
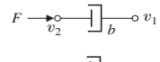
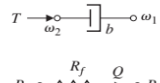
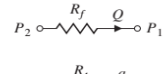
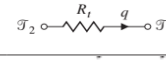
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Table 2.2 Summary of Governing Differential Equations for Ideal Elements

Type of Element	Physical Element	Governing Equation	Energy E or Power $\mathcal{P}$	Symbol
Inductive storage	Electrical inductance	$v_{21} = L\frac{di}{dt}$	$E = \frac{1}{2}Li^2$	
	Translational spring	$v_{21} = \frac{1}{k}\frac{dF}{dt}$	$E = \frac{1}{2}\frac{F^2}{k}$	
	Rotational spring	$\omega_{21} = \frac{1}{k}\frac{dT}{dt}$	$E = \frac{1}{2}\frac{T^2}{k}$	
	Fluid inertia	$P_{21} = I\frac{dQ}{dt}$	$E = \frac{1}{2}IQ^2$	
Capacitive storage	Electrical capacitance	$i = C\frac{dv_{21}}{dt}$	$E = \frac{1}{2}Cv_{21}^2$	
	Translational mass	$F = M\frac{dv_2}{dt}$	$E = \frac{1}{2}Mv_2^2$	
	Rotational mass	$T = J\frac{d\omega_2}{dt}$	$E = \frac{1}{2}J\omega_2^2$	
	Fluid capacitance	$Q = C_f\frac{dP_{21}}{dt}$	$E = \frac{1}{2}C_fP_{21}^2$	
	Thermal capacitance	$q = C_t\frac{d\mathcal{T}_2}{dt}$	$E = C_t\mathcal{T}_2$	

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Table 2.2 Summary of Governing Differential Equations for Ideal Elements

Type of Element	Physical Element	Governing Equation	Energy E or Power $\mathcal{P}$	Symbol
Energy dissipators	Electrical resistance	$i = \frac{1}{R}v_{21}$	$\mathcal{P} = \frac{1}{R}v_{21}^2$	
	Translational damper	$F = bv_{21}$	$\mathcal{P} = bv_{21}^2$	
	Rotational damper	$T = b\omega_{21}$	$\mathcal{P} = b\omega_{21}^2$	
	Fluid resistance	$Q = \frac{1}{R_f}P_{21}$	$\mathcal{P} = \frac{1}{R_f}P_{21}^2$	
	Thermal resistance	$q = \frac{1}{R_t}\mathcal{T}_{21}$	$\mathcal{P} = \frac{1}{R_t}\mathcal{T}_{21}$	

Nomenclature

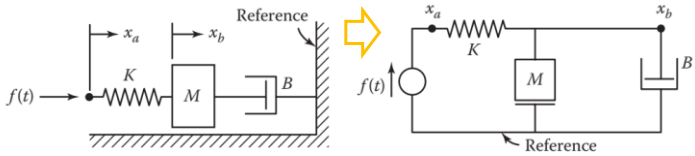
- *Through-variable:* F = force, T = torque, i = current, Q = fluid volumetric flow rate, q = heat flow rate.
- *Across-variable:* v = translational velocity, ω = angular velocity, v = voltage, P = pressure, $\mathcal{T}$ = temperature.
- *Inductive storage:* L = inductance, $1/k$ = reciprocal translational or rotational stiffness, I = fluid inertance.
- *Capacitive storage:* C = capacitance, M = mass, J = moment of inertia, C_f = fluid capacitance, C_t = thermal capacitance.
- *Energy dissipators:* R = resistance, b = viscous friction, R_f = fluid resistance, R_t = thermal resistance.

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Una alternativa al modelado físico (enfoque de red física)

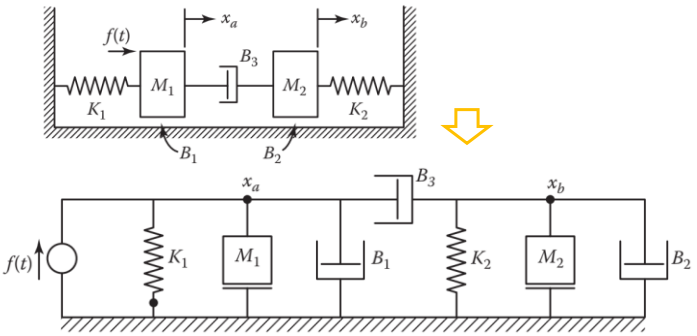
Sistemas en traslación

Se propone trazar un circuito equivalente (mecánico, térmico, fluídos, etc, similar a los circuitos eléctricos. Para el caso de sistemas mecánicos esto se logra conectando los terminales de elementos que tengan el mismo desplazamiento. Se deben plantear el equivalente a las ecuaciones de nodos de los circuitos eléctricos. En cada nodo la suma de las fuerzas concurrentes deben ser cero.



Para el nodo a $f = f_K = K(x_a - x_b)$

Para el nodo b $f_K = f_M + f_B = MD^2 x_b + BDx_b$



Para el nodo a

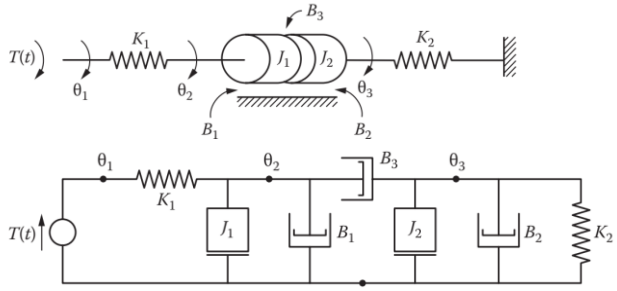
$$(M_1D^2 + B_1D + B_3D + K_1)x_a - (B_3D)x_b = f$$

Para el nodo b

$$-(B_3D)x_a + (M_2D^2 + B_2D + B_3D + K_2)x_b = 0$$

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Sistemas en rotación



Node 1: $K_1\theta_1 - K_1\theta_2 = T(t)$

Node 2: $-K_1\theta_1 + [J_1D^2 + (B_1 + B_3)D + K_1]\theta_2 - (B_3D)\theta_3 = 0$

Node 3: $-(B_3D)\theta_2 + [J_2D^2 + (B_2 + B_3)D + K_2]\theta_3 = 0$

Sistemas térmicos

TABLE 5.6
Thermal Symbols and Units

Symbol	Quantity	Metric Units
q	Rate of heat flow	Joules/second
M	Mass	Kilograms
S	Specific heat	Joules/(kilogram)(°C)
C	Thermal capacitance $C = MS$	Joules/°C
K	Thermal conductance	Joules/(second)(°C)
R	Thermal resistance	Degrees/(joule/second)
θ	Temperature	°C
h	Heat energy	Joules

La capacidad térmica determina la cantidad de calor almacenado en un cuerpo, dada una diferencia de temp.

$$h = \frac{q}{D} = C(\theta_2 - \theta_1) \quad \Rightarrow \quad q = CD(\theta_2 - \theta_1)$$

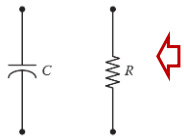
La resistencia térmica determina el flujo de calor a través de un cuerpo con diferencias de temperatura

$$q = \frac{\theta_3 - \theta_4}{R}$$

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Sistemas térmicos

Se supone que se dispone de un termómetro de mercurio con resisencia térmica R y capacidad calorífica C. El termómetro está estabilizado a una temperatura Θm y se lo sumerge en un líquido a temperatura Θo.



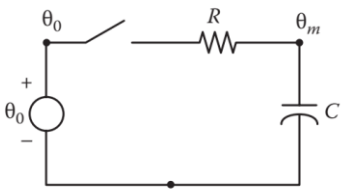
Símbolos a utilizar en el modelo

El flujo de calor desde el líquido al termómetro viene limitado por la resistencia térmica

$$q = \frac{\theta_0 - \theta_m}{R}$$

El flujo de calor entrando al Termómetro es almacenado en su capacidad térmica

$$h = \frac{q}{D} = C(\theta_m - \theta_1)$$



Combinando las ecuaciones

$$\frac{\theta_0 - \theta_m}{RD} = C(\theta_m - \theta_1)$$

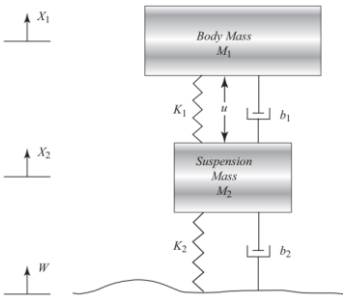
Diferenciando y ordenando

$$RC\,D\theta_m + \theta_m = \theta_0$$

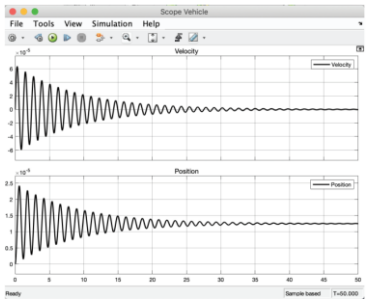
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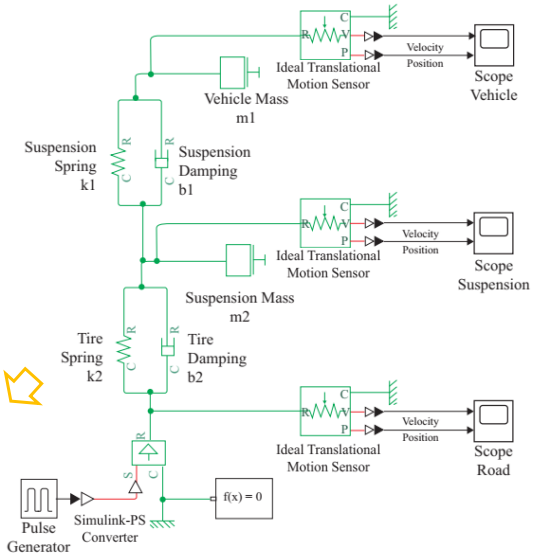
Enfoque de red física y SIMSCAPE

Sistemas en traslación: suspensión de un vehículo



X1-X2 es la variable de salida debido a que *X2-W* es difícil de medir.
U es la fuerza de un actuador pensado para restituir el sistema a su posición deseada.

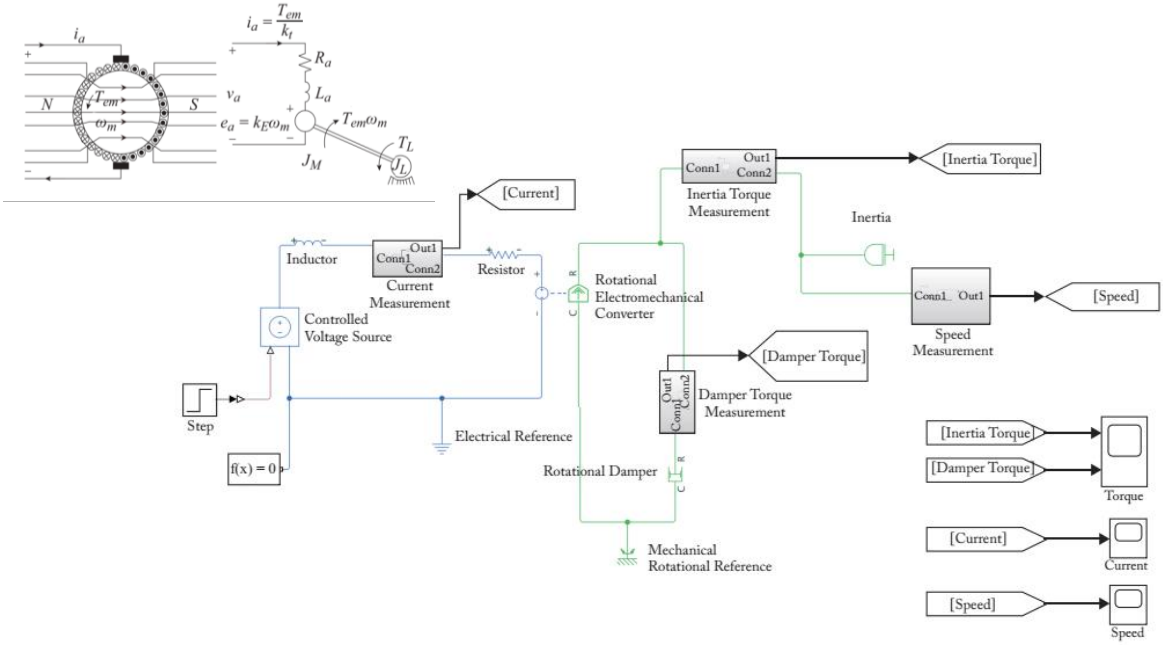




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Enfoque de red física y SIMSCAPE

Motor de DC de imán permanente

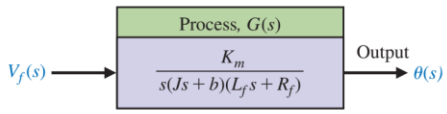


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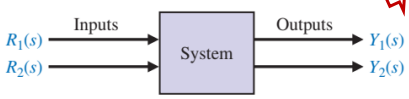
Modelos de diagramas de bloques

Consisten en la combinación de bloques que representan las funciones de transferencia de las distintas partes de un sistema. El flujo de señal es unidireccional. No se cumplen en estos diagramas las leyes de conservación.

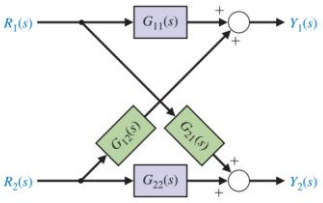
Para el caso de un motor de DC



Para el caso de un sistema de dos entradas y dos salidas



$$Y_1(s) = G_{11}(s)R_1(s) + G_{12}(s)R_2(s)$$
$$Y_2(s) = G_{21}(s)R_1(s) + G_{22}(s)R_2(s)$$

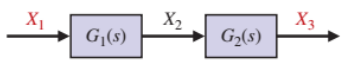
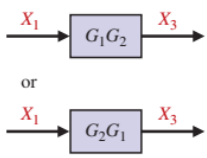
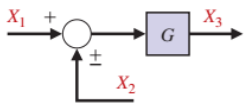
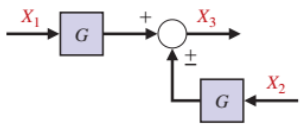
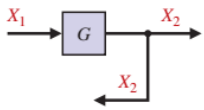
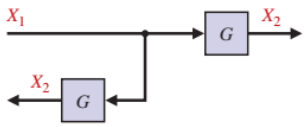


Generalizando

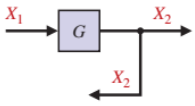
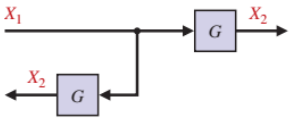
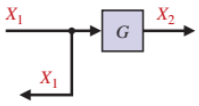
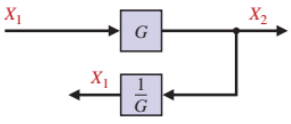
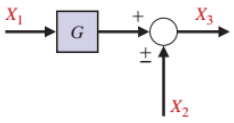
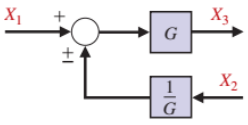
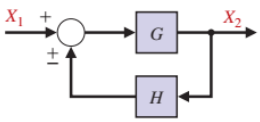
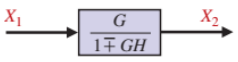
$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \\ \vdots \\ Y_I(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & \dots & G_{1J}(s) \\ G_{21}(s) & \dots & G_{2J}(s) \\ \vdots & & \vdots \\ G_{I1}(s) & \dots & G_{IJ}(s) \end{bmatrix} \begin{bmatrix} R_1(s) \\ R_2(s) \\ \vdots \\ R_J(s) \end{bmatrix}$$

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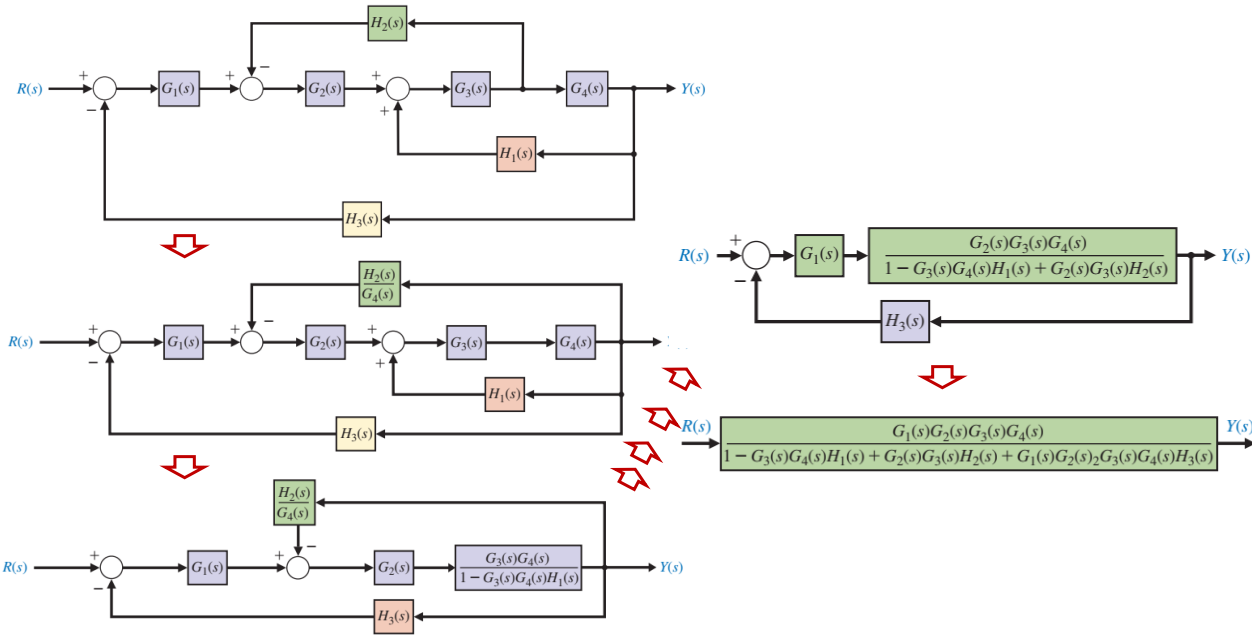
Transformaciones de diagramas de bloques

Transformation	Original Diagram	Equivalent Diagram
1. Combining blocks in cascade		
2. Moving a summing point behind a block		
3. Moving a pickoff point ahead of a block		

Transformaciones de diagramas de bloques

Transformation	Original Diagram	Equivalent Diagram
3. Moving a pickoff point ahead of a block		
4. Moving a pickoff point behind a block		
5. Moving a summing point ahead of a block		
6. Eliminating a feedback loop		

Transformaciones de diagramas de bloques. Ejemplo



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